Practical 3: Visualisation using `qplot()`

Data Science (CDA) — Academic Lab Report

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Abstract

This report documents the implementation, graphical validation, and clinical data interpretation of Practical 3 (*Visualisation using `qplot()`*) for the Data Science curriculum at ETSINF (UPV). Using the `ggplot2` framework in R, we analyze real-world surgical records from $N = 394$ patients diagnosed with keratoconus who underwent intrastromal corneal ring segment (ICRS) implantation. We evaluate corneal topography parameters (primary keratometry K_1 and orthogonal curvature K_2), post-operative outcomes ($K_{1.\text{output}}$), and surgical design choices (number of rings, incision diameter, and ring thickness). The experimental pipeline covers non-parametric local polynomial smoothing (LOESS), ordinary least squares (OLS) linear regressions, aesthetic mappings, conditional discrete distributions, multi-panel faceting, and interquartile dispersion via boxplots.

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1 Introduction and Clinical Context

Keratoconus is a progressive non-inflammatory ectatic corneal dystrophy characterized by localized thinning and biomechanical weakening of corneal collagen fibrils. This structural degeneration induces a conical protrusion of the optical dome, yielding severe irregular astigmatism, high-order visual aberrations, and significant visual acuity loss.

One prominent conservative surgical intervention consists of the placement of intrastromal corneal ring segments (ICRS). These polymethyl methacrylate (PMMA) micro-segments induce an arc-shortening mechanical effect across the corneal lamellae, producing an intentional biomechanical flattening and reshaping of the anterior curvature. The dataset `Queratocono.csv` compiles clinical pre-operative measurements, surgical procedure specifications, and post-operative evaluations for 394 treated eyes across 16 numerical and categorical features.

2 Exercise 1: Curvature Relation (K_1 vs. K_2) with Smoother

2.1 Objective and Implementation

The objective is to explore the baseline anatomical dependency between the principal meridian curvature (K_1) and its orthogonal counterpart (K_2), contrasting non-parametric local polynomial regression (LOESS) against parametric linear modeling (ordinary least squares).

```
library(ggplot2)
queratocono <- read.csv("dataset/Queratocono.csv", header = TRUE, sep = ",",
    check.names = TRUE)
```

```
# 1.a) Default non-parametric smoother (LOESS)
p1_loess <- qplot(K1, K2, data = queratocono, geom = c("point", "smooth"))

# 1.b) Parametric linear regression smoother (method = "lm")
p1_lm <- qplot(K1, K2, data = queratocono, geom = c("point", "smooth"), method
    = "lm")
```

2.2 Graphical Evaluation and Discussion

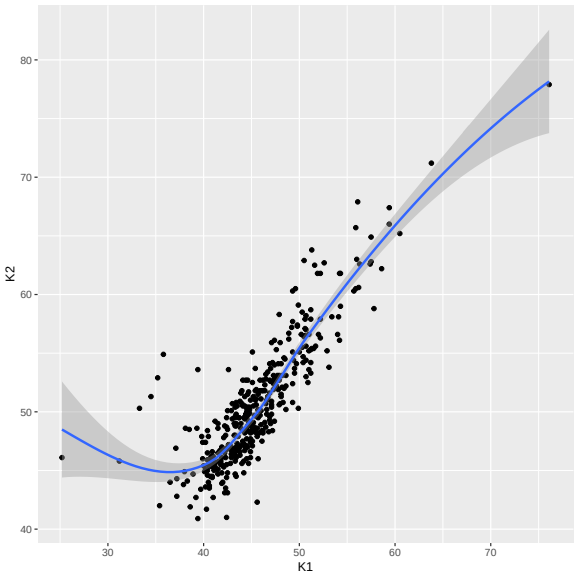


Figure 1: Default LOESS smoother on K_1 vs. K_2.

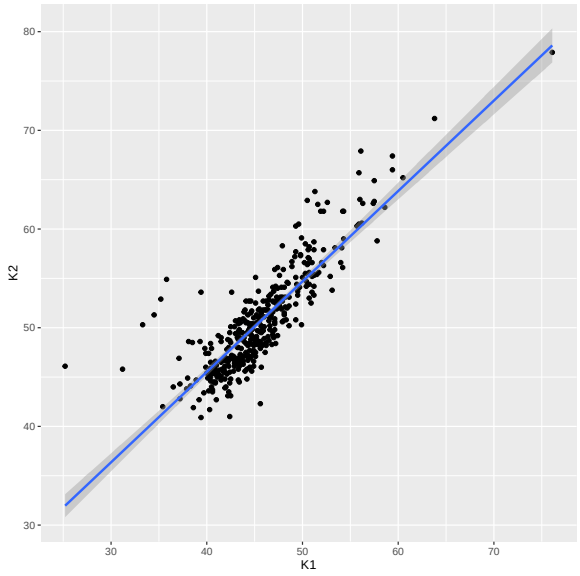


Figure 2: Linear model fit (`method = "lm"`).

The LOESS smoother (Figure 1) reveals a non-linear curvature flattening below $K_1 < 40$ diopters, transitioning into a strong linear regime across the bulk of the patient cohort ($40 \leq K_1 \leq 60$). At the extreme upper bound ($K_1 > 60$), the 95% confidence envelope widens significantly due to sample sparsity in severe advanced ectasias. Conversely, the OLS linear model (Figure 2) imposes a constant slope throughout the domain ($K_2 \approx \beta_0 + \beta_1 K_1$), confirming strong overall collinearity ($r > 0.85$) while obscuring localized deviations at low curvature boundaries.

3 Exercise 2: Keratometric Relationship Stratified by Ring Count

3.1 Objective and Implementation

We assess whether the observed linear relationship between K_1 and K_2 differs across the subgroups defined by the number of implanted rings (`na` $\in \{1, 2\}$).

```
p2 <- qplot(K1, K2,
            data = queratocono,
            colour = factor(na),
            geom = c("point", "smooth"),
            method = "lm",
            main = "Relation between K1 and K2")
```

3.2 Graphical Evaluation and Discussion

Mapping `colour = factor(na)` splits the observations and fits two independent linear models:

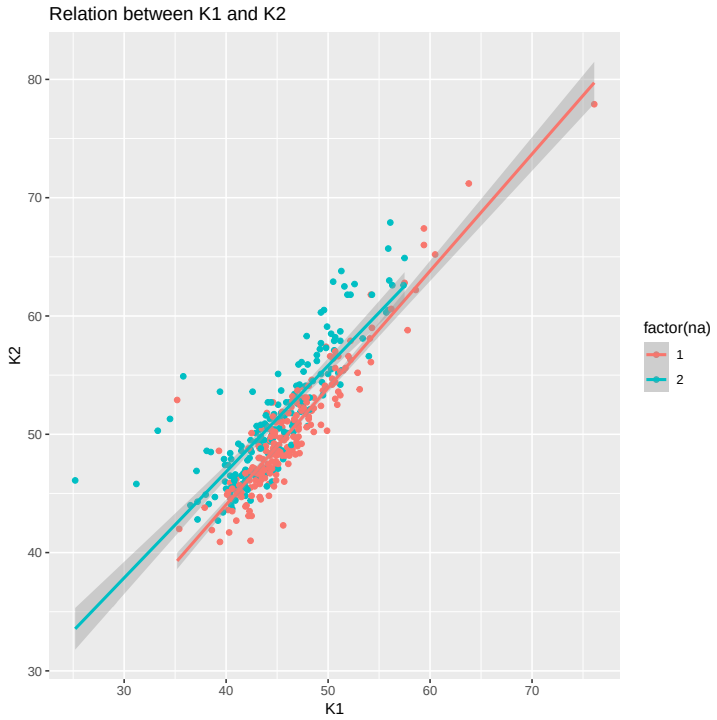


Figure 3: Stratified linear regressions for K_1 vs. K_2 by factor(`na`).

- **Single-ring group (`na = 1`):** The regression slope is slightly steeper, and points exhibit wider dispersion at higher curvature values ($K_1 > 50$).
- **Double-ring group (`na = 2`):** Observations are clustered more tightly around the fitted line, especially in the central range ($40 \leq K_1 \leq 50$), showing a marginally lower intercept and a very consistent linear correlation.

4 Exercise 3: Pre-operative vs. Post-operative Curvature

4.1 Objective and Implementation

The relationship between pre-operative curvature (K_1) and post-operative curvature ($K_{1.\text{output}}$) is displayed via a standard scatter plot.

```
p3 <- qplot(K1, K1.salida, data = queratocono, geom = "point")
```

4.2 Graphical Evaluation and Discussion

The distribution exhibits a strong positive linear correlation throughout the dataset, with the majority of values falling in the range $40 \leq K_1 \leq 55$. Two extreme data points stand out clearly:

- An isolated point in the lower-left region with low pre- and post-operative values ($K_1 \approx 25$, $K_{1.\text{output}} \approx 29$).
- An extreme outlier on the far right with $K_1 > 75$ whose post-operative measurement drops to $K_{1.\text{output}} \approx 45$, as well as an observation at $K_1 \approx 64$ with $K_{1.\text{output}} \approx 63$.

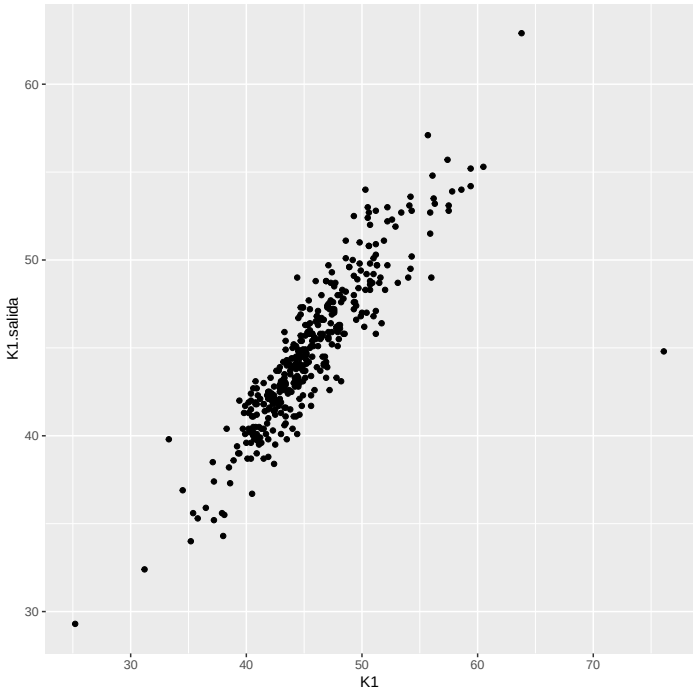


Figure 4: Scatter plot of K_1 vs. $K_{1.\text{output}}$.

5 Exercise 4: Distribution of Ring Thickness by Segment Count

5.1 Objective and Implementation

The frequency distribution of ring thickness (`grosor`) is evaluated as a discrete factor, with counts divided by ring number (`na`).

```
p4 <- qplot(factor(grosor), data = queratocono, geom = "bar", fill = factor(na))
```

5.2 Graphical Evaluation and Discussion

The sample is heavily concentrated in two specific thicknesses: 200 μm ($n = 179$) and 250 μm ($n = 127$), which together account for more than 77% of all patients. Furthermore:

- For 200 μm and 250 μm, both single-ring (`na = 1`) and double-ring (`na = 2`) interventions are frequent.
- In contrast, intermediate thicknesses such as 175 μm ($n = 10$), 225 μm ($n = 31$), and 275 μm ($n = 1$) occur exclusively in patients with two rings (`na = 2`).

6 Exercise 5: Multi-dimensional Faceting by Incision Diameter and Rings

6.1 Objective and Implementation

We examine the distribution of K_1 and K_2 segmented across a 2×2 grid defined by `diam` (rows) and `na` (columns), using ring thickness for point colour and setting an alpha transparency of 1/3.

```
p5 <- qplot(K1, K2,
            data = queratocono,
```

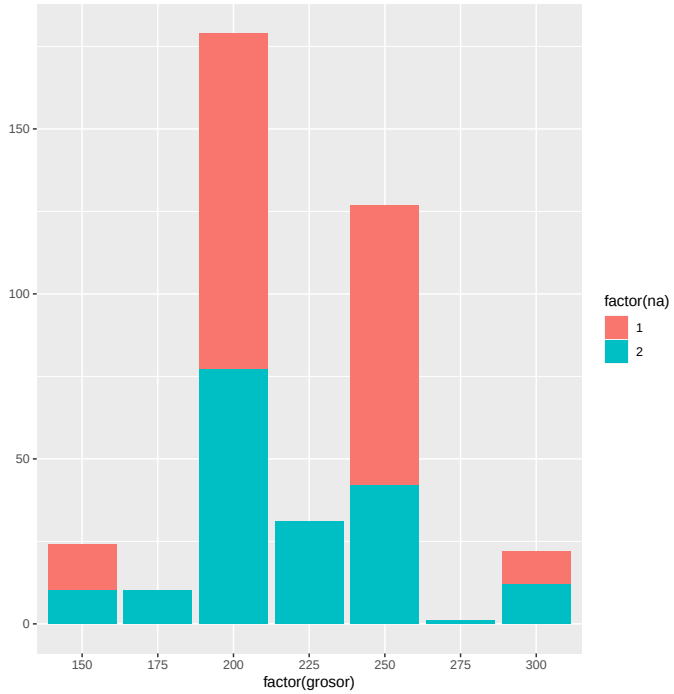


Figure 5: Stacked bar chart of factor(`grosor`) filled by factor(`na`).

```
facets = diam ~ na,
colour = factor(grosor),
alpha = I(1/3))
```

6.2 Graphical Evaluation and Discussion

The faceted layout allows direct comparison of sample density across surgical subsets:

- The panel for diam = 6 and na = 1 contains the largest volume of points and spans the widest range of K_1 values (from 35 to over 75).
- The panel for diam = 5 and na = 1 has the sparsest distribution, showing that single rings were less frequently used with a 5 mm incision diameter.
- The transparency ($\alpha = 1/3$) effectively reveals regions of high point overlap in the core interval $K_1 \in [42, 50]$, distinguishing dense clusters from isolated observations.

7 Exercise 6: Distributional Summaries via Boxplots

7.1 Objective and Implementation

Box-and-whisker plots are generated separately for K_1 and K_2 across the discrete categories of `factor(grosor)`.

```
p6_k1 <- qplot(factor(grosor), K1, data = queratocono, geom = "boxplot")
p6_k2 <- qplot(factor(grosor), K2, data = queratocono, geom = "boxplot")
```

7.2 Graphical Evaluation and Discussion

Analysis of the boxplots and underlying summary statistics highlights the following:

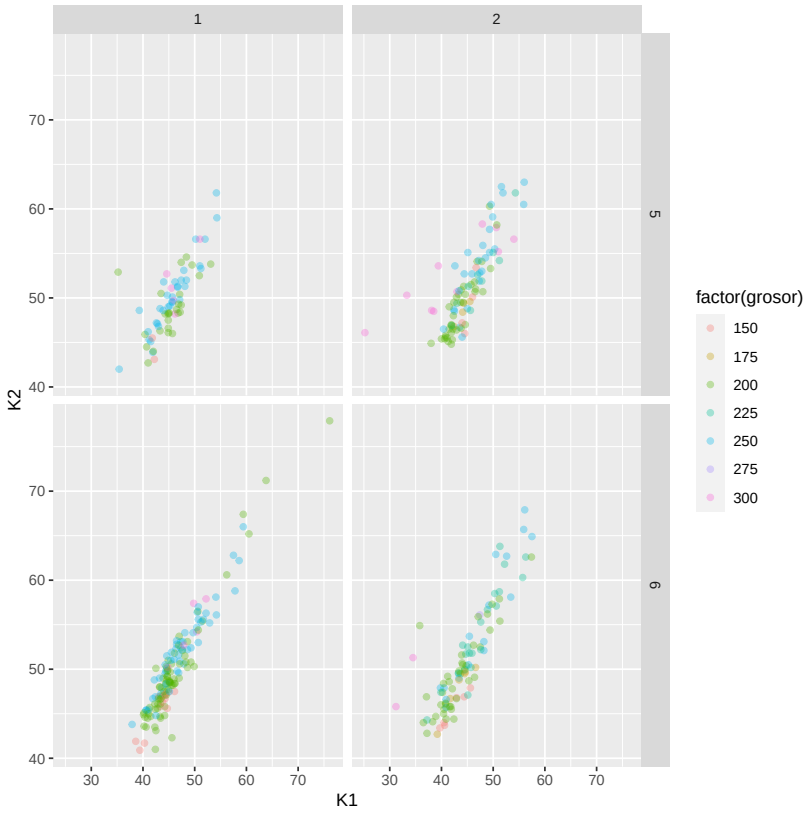


Figure 6: Faceted scatter plot (2×2 grid: `diam` $\sim$ `na`) with alpha blending.

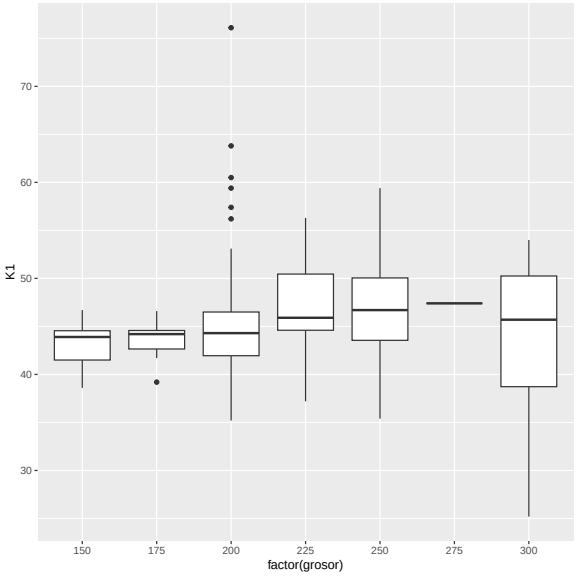


Figure 7: Boxplot of K_1 by `factor(grosor)`.

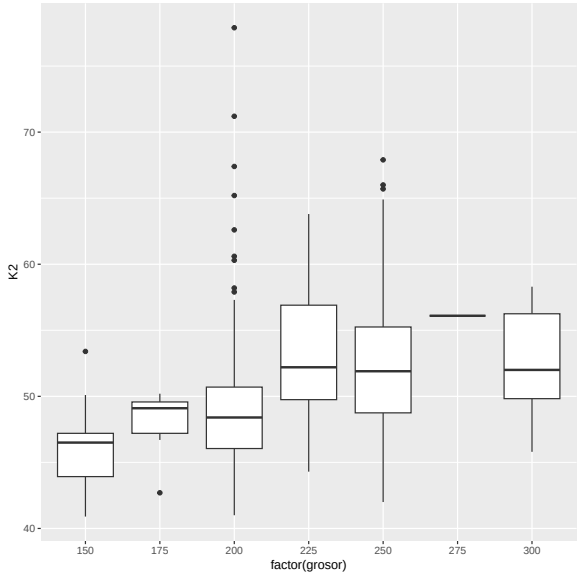


Figure 8: Boxplot of K_2 by `factor(grosor)`.

- **Median behavior is non-monotonic:** For K_1, the median rises from 43.9 in 150 μm up to 47.4 in 275 μm, but drops to 45.7 in 300 μm. For K_2, the median increases from 46.5 in 150 μm to 49.1 in 175 μm, drops slightly to 48.4 in 200 μm, and peaks at 56.1 in 275 μm before decreasing to 52.0 in 300 μm.
- **Single-observation category (275 μm):** There is only one patient recorded with a thickness of 275 μm ($N = 1$). Consequently, its boxplot collapses into a single horizontal segment representing that single measured value ($K_1 = 47.4, K_2 = 56.1$), with zero interquartile spread.
- **Outliers:** The 200 μm thickness group exhibits several prominent upper outliers in both K_1 and K_2 (reaching values above 70), reflecting the wide overall sample size ($n = 179$) present in that group.

Appendix: Complete Sequential Execution Log

The verbatim console output recorded during the sequential execution of `pract3.R` is reproduced below:

```
'data.frame': 394 obs. of  16 variables:
 $K1       : num  38.1 39.4 43.3 45.7 44.6 44.2 44.2 53.1 45.1 40.7 ...$ K2       : num
     48.6 53.6 50.4 50.1 50.4 44.8 45.8 53.8 52.5 44.5 ...
 $ch       : num  6.9 5.7 8.8 11.1 11.1 8.9 11.1 7.5 7.7 9.4 ...$ na       : int  2 2 2
     1 2 1 1 1 2 1 ...
 $Incision : int  60 140 120 30 60 180 130 60 60 30 ...$ Prof     : int  390 400 381 370
     440 327 380 400 380 477 ...
 $diam     : int  5 5 5 5 5 6 6 5 6 5 ...$ grosor   : int  300 300 200 250 200 200 200
     200 225 200 ...
 $Longitud1: int  90 105 120 160 120 150 210 160 120 160 ...$ Longitud2: int  180 210
     240 160 240 150 210 160 240 160 ...
 $grosor1  : int  300 300 200 250 200 200 200 200 200 200 ...$ grosor2  : int  300 300
     200 0 200 0 0 0 250 0 ...
 $long1    : int  90 90 120 160 120 150 210 160 90 160 ...$ long2    : int  90 120 120 0
     120 0 0 0 150 0 ...
 $K1.salida: num  35.5 42 42.7 45.8 44.4 42.8 43.1 48.7 45.7 39.8 ...$ Astig    : num
     10.1 8.9 3 1.9 2.9 2.3 3.9 2.8 3.2 3.8 ...
    K1   K2   ch na Incision Prof diam grosor Longitud1 Longitud2 grosor1
1 38.1 48.6  6.9  2       60  390    5    300        90       180     300
2 39.4 53.6  5.7  2      140  400    5    300       105       210     300
3 43.3 50.4  8.8  2      120  381    5    200       120       240     200
4 45.7 50.1 11.1  1       30  370    5    250       160       160     250
5 44.6 50.4 11.1  2       60  440    5    200       120       240     200
6 44.2 44.8  8.9  1      180  327    6    200       150       150     200
  grosor2 long1 long2 K1.salida Astig
1     300    90    90      35.5  10.1
2     300    90   120      42.0   8.9
3     200   120   120      42.7   3.0
4       0   160     0      45.8   1.9
5     200   120   120      44.4   2.9
6       0   150     0      42.8   2.3
Warning message:
‘qplot()‘ was deprecated in ggplot2 3.4.0.
‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
‘geom_smooth()‘ using method = ’loess’ and formula = ’y ~ x’
pdf
  2
Warning message:
In geom_point(method = "lm") : Ignoring unknown parameters: ‘method‘
‘geom_smooth()‘ using formula = ’y ~ x’
‘geom_smooth()‘ using formula = ’y ~ x’
pdf
  2
Warning message:
In geom_point(method = "lm") : Ignoring unknown parameters: ‘method‘
‘geom_smooth()‘ using formula = ’y ~ x’
‘geom_smooth()‘ using formula = ’y ~ x’
pdf
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```